

SPLUNK INSIDER THREAT DETECTION LAB

User & Entity Behavior Analytics (UEBA) Project

Prepared by

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6 Detection Rules	MITRE ATT&CK Mapped	Risk Score Dashboard	IR Playbooks
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1. Project Overview

This project simulates a real-world insider threat detection scenario using Splunk as the SIEM platform. It demonstrates key SOC analyst skills including log analysis, SPL query writing, UEBA-based detection, and incident response.

The project covers two insider threat actors in a fictional company (CORP.LOCAL) with 8 employees across Finance, HR, IT, Sales, Legal, and Marketing departments. The goal is to detect malicious behavior using behavioral analytics and predefined detection rules mapped to the MITRE ATT&CK framework.

1.1 What This Project Demonstrates

Skill	Where It Shows
SIEM — Splunk	SPL query writing, index management, dashboard creation
Threat Detection	6 detection rules covering multiple attack vectors
MITRE ATT&CK	8 techniques mapped across 5 tactics
UEBA / Risk Scoring	Behavioral risk model with per-user scoring
Incident Response	6 IR playbooks with triage and containment steps
Python	Synthetic log generator script (556 realistic events)

2. Project Architecture

The project has three layers working together:

2.1 Data Layer

A Python script generates 556 synthetic Windows Security Event logs in CSV format, simulating real enterprise log data. The logs include both normal user activity (400 events) and malicious insider threat activity (156 events).

Windows Event IDs used:

Event ID	Description	Used For
4624	Successful Logon	Off-hours login detection
4625	Failed Logon	Brute force / credential stuffing
4634	Logoff	Session duration tracking
4648	Explicit Credential Logon	Lateral movement detection
4663	Object Access Attempt	Sensitive file access detection
4672	Special Privileges Assigned	Privilege use tracking
4688	New Process Created	Malicious process detection
4720	User Account Created	Backdoor account detection
4728	Member Added to Group	Privilege escalation detection
6416	New External Device	USB insertion / exfiltration
7045	New Service Installed	Persistence mechanism detection
1102	Audit Log Cleared	Anti-forensics detection

2.2 Detection Layer

Six SPL (Search Processing Language) detection rules written in Splunk that identify malicious patterns in the log data. Each rule is saved as both a Splunk Report and an Alert.

2.3 Presentation Layer

A Splunk dashboard with 6 panels visualizing the detections, including a UEBA risk scoring leaderboard showing which users have the highest anomaly scores.

3. Insider Threat Scenarios

The project simulates two distinct insider threat actors with different motivations and attack methods.

3.1 Threat Actor 1 — sara.kim (Sales Associate)

Motivation: Data theft — copying confidential client and salary data to a personal USB drive before leaving the company.

Phase	Week	Activity	Event ID
1	Week 1	Logs in after midnight repeatedly — reconnaissance	4624 + OffHoursLogin flag
2	Week 2	Accesses salary, payroll, and merger files in bulk	4663 + Confidential files
3	Week 3	Inserts USB drive — SanDisk 16GB detected	6416 — RemovableStorage
4	Week 3	Attempts to log in as other employees — fails	4625 — multiple accounts
5	Week 3	Tries to add herself to Domain Admins group	4728 — Domain Admins

3.2 Threat Actor 2 — ravi.kumar (IT SysAdmin)

Motivation: Credential theft and sabotage — abusing IT admin privileges to dump credentials, move laterally, establish persistence, and cover tracks.

Phase	Week	Activity	Event ID
1	Week 1	RDP sessions at odd hours from IT workstation	4624 + RDP + OffHours
2	Week 2	Executes Mimikatz — dumps LSASS credentials	4688 — mimikatz.exe
3	Week 2	Uses PsExec to move laterally to other machines	4648 — explicit credentials
4	Week 3	Installs hidden Windows service for persistence	7045 — new service
5	Week 3	CLEARs the audit log to hide all evidence	1102 — log cleared
6	Week 3	Creates backdoor admin account svc_backup2	4720 — new account

4. SPL Detection Rules

All 6 detection rules are saved as Splunk Reports and Alerts. Each is also stored as a .spl file in the detections/ folder of this repository.

Detection 1 — Off-Hours Login

File: detections/offhours_login.spl

The screenshot shows the Splunk Enterprise interface with a new search being created. The search query is:

```
index="windows" EventID=4624 AdditionalInfo="*OffHoursLogin=True*"
| stats count by Username, Computer
| where count > 3
| sort -count
```

The search results show 27 events. The table below displays the results:

Username	Computer	count
CORP.LOCAL\sara.kim	WS-SAL-004	15
CORP.LOCAL\ravi.kumar	WS-IT-007	12

Detects users logging in outside 8AM-6PM more than 3 times. Flags sara.kim and ravi.kumar.

Detection 2 — Bulk Sensitive File Access

File: detections/bulk_file_access.spl

The screenshot shows the Splunk Enterprise interface with a new search being created. The search query is:

```
index="windows" EventID=4663
| search TargetObject="*Confidential*" OR TargetObject="*Salary*"
OR TargetObject="*Restricted*" OR TargetObject="*Payroll*"
| stats count by Username, TargetObject
| where count > 5
| sort -count
```

The search results show 32 events. The table below displays the results:

Username	TargetObject	count
CORP.LOCAL\sara.kim	C:\Finance\Payroll_Oct2024.xlsx	8
CORP.LOCAL\sara.kim	C:\HR\Employee_Salary_Data.csv	8
CORP.LOCAL\sara.kim	C:\Legal\Contracts_Confidential.docx	6

Detects bulk access to files tagged as sensitive. Identifies users reading payroll, merger, and salary data repeatedly.

Detection 3 — USB Device Insertion

File: detections/usb_insertion.spl

The screenshot shows the Splunk Enterprise interface with a search query for USB device insertion events. The search bar contains the following query:

```
index="windows" EventID=6416
| table Timestamp, Username, Computer, TargetObject, AdditionalInfo
| sort Timestamp
```

The search results show 8 events. The table below represents the data shown in the search results:

Timestamp	Username	Computer	TargetObject	AdditionalInfo
2024-10-06 22:40:27	CORP.LOCAL\sara.kim	WS-SAL-004	USB\VID_1307&PID_0165\KINGSTON_32GB	DeviceType=RemovableStorage CopyDetected=True
2024-10-11 03:09:45	CORP.LOCAL\sara.kim	WS-SAL-004	USB\VID_090C&PID_1000\2024_SANDISK_16GB	DeviceType=RemovableStorage CopyDetected=True
2024-10-11 22:34:10	CORP.LOCAL\sara.kim	WS-SAL-004	USB\VID_1307&PID_0165\KINGSTON_32GB	DeviceType=RemovableStorage CopyDetected=True
2024-10-19 03:51:03	CORP.LOCAL\sara.kim	WS-SAL-004	USB\VID_090C&PID_1000\2024_SANDISK_16GB	DeviceType=RemovableStorage CopyDetected=True
2024-10-19 06:31:09	CORP.LOCAL\sara.kim	WS-SAL-004	USB\VID_058F&PID_6387\GENERIC_USB_8GB	DeviceType=RemovableStorage CopyDetected=True
2024-10-21	CORP.LOCAL\sara.kim	WS-SAL-004	USB\VID_1307&PID_0165\KINGSTON_32GB	DeviceType=RemovableStorage

Captures every removable storage device insertion. sara.kim is the only user with USB events, making her immediately suspicious.

Detection 4 — Malicious Process Execution

File: detections/malicious_process.spl

The screenshot shows the Splunk Enterprise interface with a search query for malicious process execution. The search bar contains the following query:

```
index="windows" EventID=4688
| search ProcessName="mimikatz.exe" OR ProcessName="psexec.exe"
OR ProcessName="wmic.exe"
| stats count by Username, ProcessName, Computer
| sort -count
```

The search results show 15 events. The table below represents the data shown in the search results:

Username	ProcessName	Computer	count
CORP.LOCAL\ravi.kumar	mimikatz.exe	WS-IT-007	5
CORP.LOCAL\ravi.kumar	psexec.exe	WS-IT-007	5
CORP.LOCAL\ravi.kumar	wmic.exe	WS-IT-007	5

Fires immediately on any execution of known credential dumping or lateral movement tools. These are almost never legitimate.

Detection 5 — Audit Log Cleared

File: detections/audit_log_cleared.spl

The screenshot shows the Splunk Enterprise interface. The search bar contains the query: `index="windows" EventID=1102`. Below the search bar, the results are displayed in a table view. The table has columns: Timestamp, Username, Computer, and AdditionalInfo. There are 3 events listed, all with the same username: CORP.LOCAL\ravi.kumar. The AdditionalInfo column shows: SubjectUserName=ravi.kumar | AuditLogCleared=True.

Timestamp	Username	Computer	AdditionalInfo
2024-10-09 13:20:28	CORP.LOCAL\ravi.kumar	DC01	SubjectUserName=ravi.kumar AuditLogCleared=True
2024-10-11 12:16:54	CORP.LOCAL\ravi.kumar	DC01	SubjectUserName=ravi.kumar AuditLogCleared=True
2024-10-14 17:36:03	CORP.LOCAL\ravi.kumar	DC01	SubjectUserName=ravi.kumar AuditLogCleared=True

Critical P1 alert. Any occurrence of this event means someone is trying to cover their tracks. Triggered by ravi.kumar.

Detection 6 — User Risk Score (UEBA)

File: detections/user_risk_score.spl

The screenshot shows the Splunk Enterprise interface. The search bar contains the query: `index="windows" | eval RiskScore=case(Severity="Critical", 10, Severity="High", 5, Severity="Medium", 3, Severity="Low", 1, true(), 0) | stats sum(RiskScore) as TotalRisk, count as TotalEvents by Username | sort -TotalRisk`. Below the search bar, the results are displayed in a table view. The table has columns: Username, TotalRisk, and TotalEvents. There are 6 users listed, sorted by TotalRisk in descending order.

Username	TotalRisk	TotalEvents
CORP.LOCAL\ravi.kumar	526	76
CORP.LOCAL\sara.kim	395	80
CORP.LOCAL\emily.brown	10	74
CORP.LOCAL\mike.chen	8	70
CORP.LOCAL\anita.patel	7	66
CORP.LOCAL\john.smith	6	66

Aggregates all events per user and assigns a cumulative risk score based on severity. Produces a leaderboard that instantly shows who the highest risk users are.

5. MITRE ATT&CK Mapping

All detections in this project are mapped to the MITRE ATT&CK Enterprise framework, the industry standard for categorizing adversary tactics and techniques.

Detection	MITRE Technique	Tactic	Threat Actor
Off-Hours Login	T1078 — Valid Accounts	Initial Access	sara.kim, ravi.kumar
Bulk File Access	T1005 — Data from Local System	Collection	sara.kim, ravi.kumar
USB Exfiltration	T1052.001 — Physical Medium	Exfiltration	sara.kim
Credential Dump	T1003.001 — LSASS Memory	Credential Access	ravi.kumar
Lateral Movement	T1550.002 — Pass the Hash	Lateral Movement	ravi.kumar
Log Cleared	T1070.001 — Clear Event Logs	Defense Evasion	ravi.kumar
New Service	T1543.003 — Windows Service	Persistence	ravi.kumar
Backdoor Account	T1136.001 — Local Account	Persistence	ravi.kumar

6. Splunk Dashboard

The dashboard Insider Threat Detection - UEBA contains 6 panels built from the detection queries above.

#	Panel Name	Visualization	What It Shows
1	User Risk Leaderboard	Bar Chart	Top users by cumulative risk score
2	Off-Hours Login Timeline	Line Chart	Daily off-hours login count per user
3	Sensitive File Access	Statistics Table	Which users accessed which sensitive files
4	USB Device Insertions	Statistics Table	All removable storage events with timestamps
5	Malicious Processes	Statistics Table	Credential dumping and hacking tool executions
6	Audit Logs Cleared	Single Value	Count of anti-forensics events detected

Dashboard screenshots are available in the `dashboards/screenshots/` folder of this repository.

7. How to Run This Lab

Prerequisites

- Splunk Enterprise (Free Trial) — splunk.com/download
- Python 3.x (to re-generate logs if needed)
- Git (to clone the repository)

Step 1 — Clone the Repository

```
git clone https://github.com/baddilasowmya/splunk-insider-threat-detection
```

```
cd splunk insider-threat Detection
```

Step 2 — Start Splunk and Upload Logs

1. Open Splunk at <http://localhost:8000>
2. Go to Settings > Add Data > Upload
3. Select data/insider_threat_logs.csv
4. Set source type to csv
5. Create a new index named windows
6. Click Submit

Step 3 — Run Detection Queries

7. Open Search & Reporting app in Splunk
8. Copy each .spl file from detections/ folder
9. Paste into the search bar and run
10. Save each as a Report and Alert

Step 4 — Import Dashboard

11. In Splunk go to Dashboards > Create New Dashboard
12. Click Edit Source and paste the XML from dashboards/insider_threat_dashboard.xml
13. Save — all 6 panels will load automatically

8. Tools & Technologies

Tool	Version/Type	Purpose
Splunk Enterprise	Free Trial / 9.x	SIEM platform, SPL queries, dashboards

Python 3	3.x	Synthetic log generation script
Windows Event Logs	Security / System	Log format and Event ID reference
MITRE ATT&CK	Enterprise v14	Detection technique mapping
Git / GitHub	Latest	Version control and portfolio hosting

9. Author

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